Introduction and Statistics Review



Teaching Team

Introduction and Statistics Review

Course Description

This course provides an introduction to econometrics theory and practice. We begin with a comprehensive discussion of simple and multiple regression analysis. More advanced topics will be tackled later in the semester, such as time series analysis, panel data, and binary models. The purpose of the course is to introduce you the statistical methods that economists use to test economic theory and interpret economic data, and to give you experience in estimating econometric models with real world data.

This course has a strong applied emphasis. We will use STATA for data analysis. The course will provide you the opportunity to establish your skills of applying STATA in empirical study. Instructions on how to use the software will be given in tutorials and, if necessary, some of the tutorials may take place in the computer lab.

Course Objective

By the end of this course, you should be able to:

1. Be able to interpret empirical studies in ways that are interesting, enlightening, and useful
2. Be aware of potential problems with the methods and know how to make corrections if these problems are present
3. Formulate your own econometric models to perform empirical investigations in business economics and finance
4. Collect and organize economic and financial data, and use statistical software packages (STATA) for empirical analysis

Textbook and Software

Textbook:

- Introductory Econometrics: A Modern Approach, Wooldridge, Jeffrey M., Cengage Learning, 6th edition, 2015
- Required readings and practice questions.

Software: STATA

The statistic software STATA will be used extensively in the course

- Class examples are illustrated using STATA
- Programming code for homework will be provided only for STATA users
- TA STATA tutorials

STATA is available at the University computer lab (https://itso.cuhk.edu.cn/taxonomy/term/164)

Or you can download it at (http://cuhko365.sharepoint.com/sites/swcenter)

One good introduction can be found at http://data.princeton.edu/stata/

Evaluation

- **Four assignments (25%)**

-- Lowest score will be dropped, i.e. only three of them are counted

- **Midterm exam (30%)**

-- In class, details to be announced later

-- In case of illness or accident at the time of the midterm - with proper documentation from a doctor or the police, the final will be weighted 75%.

- **Final exam (45%)**

-- Final examination will be cumulative, but will put slightly more emphasize on the materials after the midterm exam

- **Attendance**

-- While attendance itself does not account for any points, repeated failures to attend the class will subject to up to 5 points punishment

Supportings

- **Recommended end-of-chapter problems.**

Each week, recommended problems will be posted on the course webpage. Those problems are for practice purpose and, therefore, they won't be marked. Solutions will be posted after tutorials.

- **TA Tutorials**

Eight tutorials will be provided by TA. The purpose of these tutorials is to help students use STATA and provide extra help to solve recommended end-of-chapter problems.

- **Office Hours:**

You may stop by during my office hours (or the TAs') with any questions or by appointment.

Emails are not preferred since it is hard to type equations in emails

Course Website

The course website (Blackboard) provides valuable information related to the course. On the website you will find:

- Lecture slides
- Additional reading materials
- Data and programming code
- Suggested problems and solution
- Assignments and solution
- Announcements
- Discussion page

You should check it regularly as you are responsible for any information posted there in addition to any announcements made in class

Tentative Schedule

Week	Content	Chapter
1	Introduction of Econometrics and Statistics Review	1/Appendix
2	Simple Linear Regression Model I	2
3	Multiple Linear Regression Model I: Estimation	3
4	Multiple Linear Regression Model II: Estimation	3
5	Multiple Linear Regression Model III: Inference	4
6	Multiple Linear Regression Model IV: Inference	4
7	Multiple Linear Regression Model IV: Further issues	5/6
8	Midterm	
9	Instrumental Variables	15
10	Binary Variables	7
11	Heteroskedasticity	8
12	Panel Data Analysis I	13
13	Panel Data Analysis I	14
14	Carrying Out an Empirical Project	19
	Final Exam	

How to do a good job in this course?

- Attend classes
- Read textbook
- Finish homework independently
- Do recommended practice questions, including computer exercise
- Ask questions

Is This Course Useful? YES!

- Reason #1: It makes you competitive when you want to be a RA for SME professors
- Reason #2: It is very helpful when you apply for master programs in Finance, Economics, Business Analytics, . . .
- Reason #3: It is a necessary course if you want to write an empirical paper
- Reason #4: It is important if you want to find a job in data analysis: Amazon, Alibaba, Tencent, JD, DiDi, investment banks, . . . (Big Data!!!)
- …

Any Questions?

What is Econometrics?

"There are several aspects of the quantitative approach to economics, and no single one of these aspects, taken by itself, should be confounded with econometrics. Thus, econometrics is by no means the same as economic statistics. Nor is it identical with what we call general economic theory, although a considerable portion of this theory has a definitely quantitative character. Nor should econometrics be taken as synonymous with the application of mathematics to economics. Experience has shown that each of these three view-points, that of **statistics, economic theory, and mathematics**, is a necessary, but not by itself a sufficient, condition for a real understanding of the quantitative relations in modern economic life. It is the unification of all three that is powerful. And it is this unification that constitutes econometrics."

— Ragnar Frisch, *Econometrica*, (1933), 1, pp. 1-2.

The Nature of Econometrics and Economic Data

- What is econometrics?
  - Econometrics = use of statistical methods to analyze economic data
  - Econometricians typically analyze nonexperimental data

- Typical goals of econometric analysis
  - Estimating relationships between economic variables
  - Testing economic theories and hypotheses
  - Forecasting economic variables
  - Evaluating and implementing government and business policy

Steps in an empirical analysis

In economics, theory and empirical analysis are both important. An **empirical analysis** uses data to test a theory, estimate an economic relationship, or determine the effects of a policy or intervention.

Step 1: Be very precise in posing the question you hope to answer.

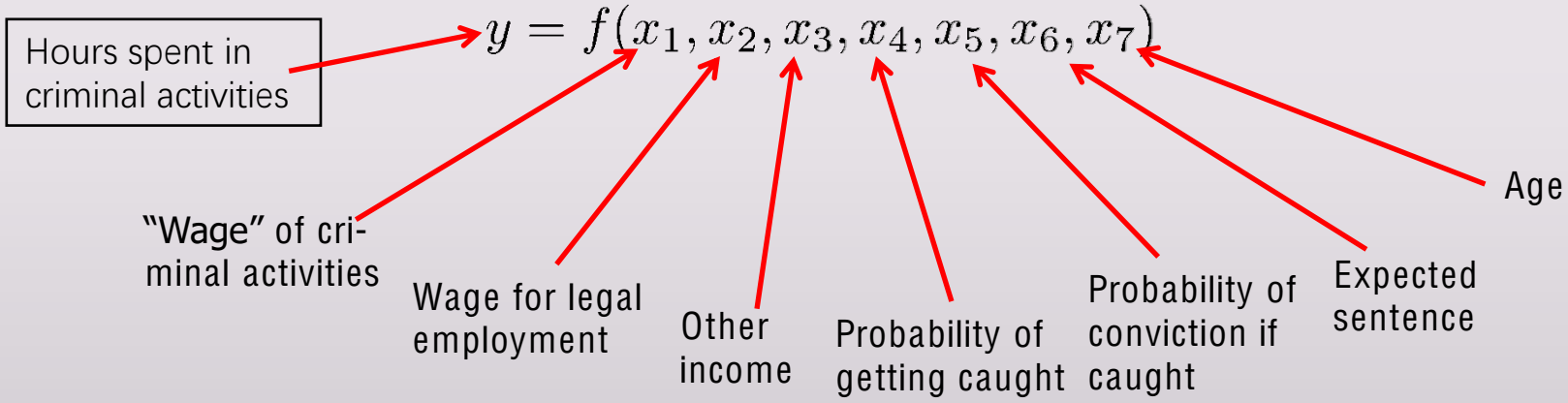
- Does the education increase one's salary?
- What affects a person's decision to participate in crime?

Step 2: Specify an **economic model**, or at least a conceptual model, to study the phenomenon of interest. For example,

- Maybe micro- or macromodels
- Often use optimizing behaviour, equilibrium modeling, …
- Establish relationships between economic variables
- Examples: demand equations, pricing equations, …

Steps in an empirical analysis

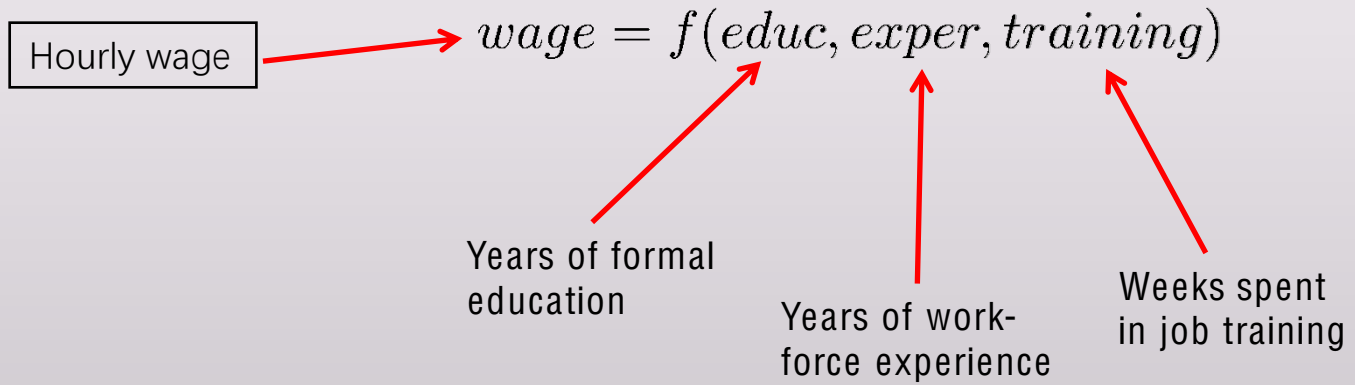
- Economic model of crime (Becker (1968))
  - Derives equation for criminal activity based on utility maximization



  - Functional form of relationship not specified
  - Equation could have been postulated without economic modeling

Steps in an empirical analysis

- Model of job training and worker productivity
  - What is effect of additional training on worker productivity?
  - Formal economic theory not really needed to derive equation:



  - Other factors may be relevant, but these are the most important (?)

Steps in an empirical analysis

Step 3: Turn the economic model into an **econometric model.**

Our focus is on econometric models. Here is where we resolve certain difficulties and ambiguities concerning an economic model. For example, in a study of criminal behavior:

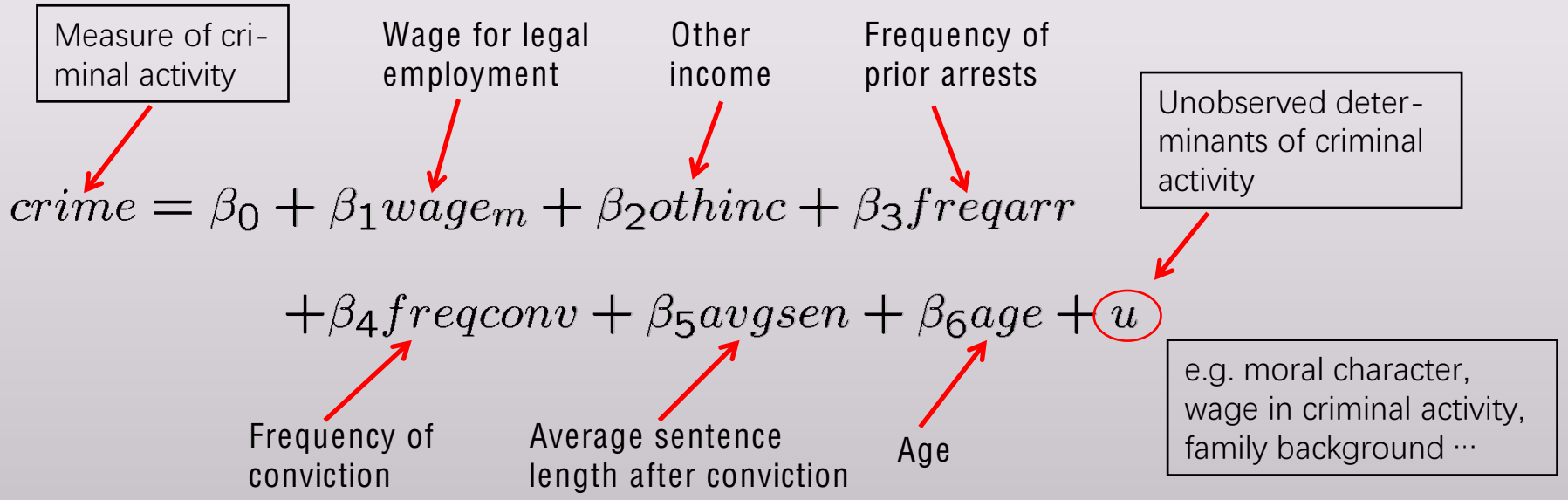
(1) What is the exact functional relationship among economic variables?

(2) How should we measure the probability of being caught committing a crime?

(3) How do we account for unobserved factors that make relationships among variables inexact?

Steps in an empirical analysis

- Econometric model of criminal activity
  - The functional form has to be specified
  - Variables may have to be approximated by other quantities



Steps in an empirical analysis

- Econometric model of job training and worker productivity

$$wage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 training + u$$

Hourly wage

Years of formal education

Years of work-force experience

Weeks spent in job training

Unobserved deter-minants of the wage

e.g. innate ability, quality of education, family background ···

- Most of econometrics deals with the specification of the error
- Econometric models may be used for hypothesis testing u
  - For example, the parameter β_3 represents "effect of training on wage"
  - How large is this effect? Is it different from zero?

Steps in an empirical analysis

- Step 1: Carefully pose a question.
- Step 2: Specify an economic or conceptual model.
- Step 3: Turn the economic model into an econometric model.
- Step 4: Collect data on the variables and use statistical methods to estimate the parameters, construct confidence intervals for the parameters, and test hypotheses

Data and Data Set

- Data are the facts and figures collected, analyzed, and summarized for presentation and interpretation.
- All the data collected in a particular study are referred to as the data set for the study.

Elements, Variables, and Observations

- **Elements** are the entities on which data are collected.
- A **variable** is a characteristic of interest for the elements.
- The set of measurements obtained for a particular element is called an **observation**. Elements, Variables, and Observations
- A data set with n elements contains n observations.
- The total number of data values in a complete data set is the number of elements multiplied by the number of variables

The structure of economic data

- Econometric analysis requires data
- Different kinds of economic data sets
  - Cross-sectional data
  - Time series data
  - Pooled cross sections
  - Panel/Longitudinal data
- Econometric methods depend on the nature of the data used
  - Use of inappropriate methods may lead to misleading results

The structure of economic data

- Cross-sectional data sets
  - Sample of individuals, households, firms, cities, states, countries, or other units of interest at a given point of time/in a given period
  - Cross-sectional observations are more or less independent
  - For example, **pure random sampling** from a population
  - Sometimes pure random sampling is violated, e.g. units refuse to respond in surveys, or if sampling is characterized by clustering
  - Cross-sectional data typically encountered in applied microeconomics

The structure of economic data

Cross-sectional data sets

- In this course, we will assume that a cross-sectional data set represents a random sample. That is, each unit in the population has the same chance of appearing the sample
- Random sampling generates observations that are independent and identically distributed.
- Intuitively, a random sample is representative of the population of interest, and gives us the best chance of learning about the population.

Example: Class attendance and final exam score

Cross-sectional data sets

- Cross-sectional data set on wages and other characteristics

TABLE 1.1 A Cross-Sectional Data Set on Wages and Other Individual Characteristics

obsno	wage	educ	exper	female	married
1	3.10	11	2	1	0
2	3.24	12	22	1	1
3	3.00	11	2	0	0
4	6.00	8	44	0	1
5	5.30	12	7	0	1
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
525	11.56	16	5	0	1
526	3.50	14	5	1	0

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Indicator variables (1 = yes, 0 = no)

Observation number

Hourly wage

Years of education

Years of experience

Cross-sectional data sets

- Cross-sectional data on growth rates and country characteristics

TABLE 1.2 A Data Set on Economic Growth Rates and Country Characteristics

obsno	country	gpcrgdp	govcons60	second60
1	Argentina	0.89	9	32
2	Austria	3.32	16	50
3	Belgium	2.56	13	69
4	Bolivia	1.24	18	12
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
61	Zimbabwe	2.30	17	6

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Average growth rate of real per capita GDP

Government consumption as a percentage of GDP

Adult secondary education rates

The structure of economic data

- Time series data
  - Observations of a variable or several variables over time
  - For example, stock prices, money supply, consumer price index, gross domestic product, annual homicide rates, automobile sales, …
  - Time series observations are typically **serially correlated**
  - Ordering of observations conveys important information
  - Data frequency: daily, weekly, monthly, quarterly, annually, …
  - Typical features of time series: **trends and seasonality**
  - Typical applications: applied macroeconomics and finance
  - Example: daily stock price

Time series data

- Time series data on minimum wages and related variables

TABLE 1.3 Minimum Wage, Unemployment, and Related Data for Puerto Rico

obsno	year	avgmin	avgcov	prunemp	prgnp
1	1950	0.20	20.1	15.4	878.7
2	1951	0.21	20.7	16.0	925.0
3	1952	0.23	22.6	14.8	1015.9
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
37	1986	3.35	58.1	18.9	4281.6
38	1987	3.35	58.2	16.8	4496.7

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Average minimum wage for the given year

Average coverage rate

Unemployment rate

Gross national product

The structure of economic data

- Pooled cross sections
  - Two or more cross sections are combined in one data set
  - Cross sections are drawn **independently** of each other
  - Pooled cross sections often used to evaluate policy changes
  - Example:
    - Evaluate effect of change in property taxes on house prices
    - Random sample of house prices for the year 1993
    - A new random sample of house prices for the year 1995
    - Compare before/after (1993: before reform, 1995: after reform)

Pooled cross sections

- Pooled cross sections on housing prices

TABLE 1.4 Pooled Cross Sections: Two Years of Housing Prices

obsno	year	hprice	proptax	sqrft	bdrms	bthrms
1	1993	85,500	42	1600	3	2.0
2	1993	67,300	36	1440	3	2.5
3	1993	134,000	38	2000	4	2.5
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
250	1993	243,600	41	2600	4	3.0
251	1995	65,000	16	1250	2	1.0
252	1995	182,400	20	2200	4	2.0
253	1995	97,500	15	1540	3	2.0
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
520	1995	57,200	16	1100	2	1.5

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Property tax

Size of house in square feet

Number of bedrooms

Number of bathrooms

Before reform

After reform

The structure of economic data

- Panel or longitudinal data
  - The same cross-sectional units are followed over time
  - Panel data have a cross-sectional and a time series dimension
  - Panel data can be used to account for time-invariant unobservables
  - Panel data can be used to model lagged responses
  - Example:
    - City crime statistics; each city is observed in two years
    - Time-invariant unobserved city characteristics may be modeled
    - Effect of police on crime rates may exhibit time lag

Panel Data

- Two-year panel data on city crime statistics

TABLE 1.5 A Two-Year Panel Data Set on City Crime Statistics

obsno	city	year	murders	population	unem	police
1	1	1986	5	350,000	8.7	440
2	1	1990	8	359,200	7.2	471
3	2	1986	2	64,300	5.4	75
4	2	1990	1	65,100	5.5	75
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
297	149	1986	10	260,700	9.6	286
298	149	1990	6	245,000	9.8	334
299	150	1986	25	543,000	4.3	520
300	150	1990	32	546,200	5.2	493

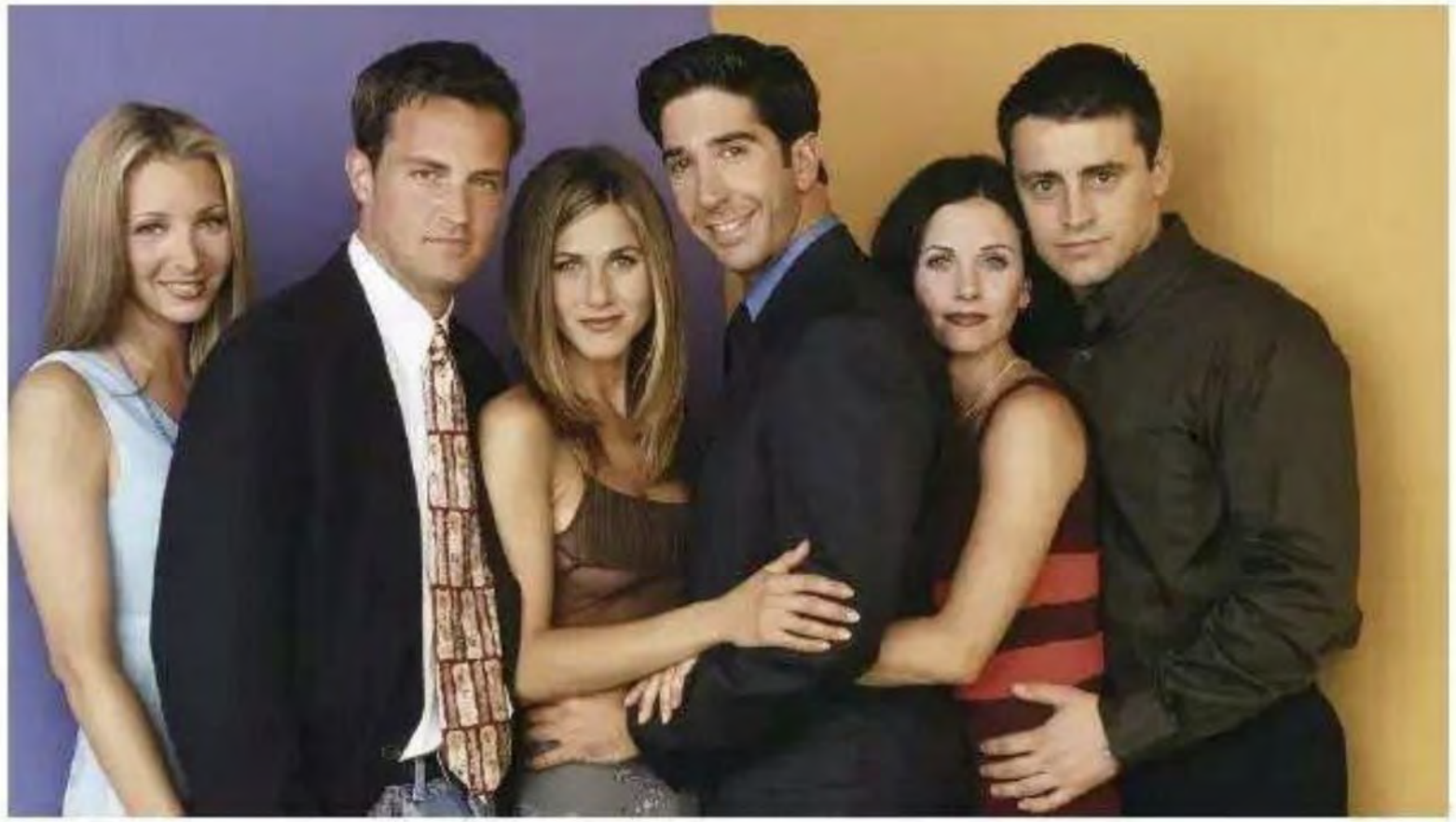
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Each city has two time series observations

Number of police in 1986

Number of police in 1990

The structure of economic data



2003

The structure of economic data



The structure of economic data



1993



2003



2013

The structure of economic data



Causality and the notion of ceteris paribus

The concept of causality is key in econometrics

In most tests of economic theory, the goal is to infer that on variable has a causal effect on another variable,

- How can we know that more class attendance causes better student performance on final exam (on average)?
- How can we know another year of education causes an increase in wages (on average)?

confounding factor ⇒ conditional

Causality and the notion of ceteris paribus

- Causality and the notion of ceteris paribus

Definition of causal effect of x on y:

> "How does variable y change if variable x is changed but all other relevant factors are held constant"

- Most economic questions are ceteris paribus questions
- It is important to define which causal effect one is interested in
- It is useful to describe how an experiment would have to be designed to infer the causal effect in question

Causality and the notion of ceteris paribus

- Causal effect of fertilizer on crop yield
  - “By how much will the production of soybeans increase if one increases the amount of fertilizer applied to the ground”
  - What are the potential ‘‘confounding factors‘‘?
- What is the ideal experiment?

Causality and the notion of ceteris paribus

- Causal effect of fertilizer on crop yield
  - "By how much will the production of soybeans increase if one increases the amount of fertilizer applied to the ground"
  - Implicit assumption: all other factors that influence crop yield such as quality of land, rainfall, presence of parasites etc. are held fixed
- Experiment:
  - Choose several one-acre plots of land; randomly assign different amounts of fertilizer to the different plots; compare yields
  - Experiment works because amount of fertilizer applied is unrelated to other factors influencing crop yields

Causality and the notion of ceteris paribus

- Measuring the return to education
  - “If a person is chosen from the population and given another year of education, by how much will his or her wage increase?”
  - What are the potential ‘‘confounding factors‘‘?
- What is the ideal experiment?

Causality and the notion of ceteris paribus

- Measuring the return to education
  - "If a person is chosen from the population and given another year of education, by how much will his or her wage increase?"
  - Implicit assumption: all other factors that influence wages such as experience, family background, intelligence etc. are held fixed
- Experiment:
  - Choose a group of people; randomly assign different amounts of education to them (infeasible!); compare wage outcomes
  - Problem without random assignment: amount of education is related to other factors that influence wages (e.g. intelligence)

Causality and the notion of ceteris paribus

- Effect of law enforcement on city crime level
  - “If a city is randomly chosen and given ten additional police officers, by how much would its crime rate fall?”
  - What are the potential ‘‘confounding factors‘‘?
- What is the ideal experiment?

Causality and the notion of ceteris paribus

- Effect of law enforcement on city crime level
  - “If a city is randomly chosen and given ten additional police officers, by how much would its crime rate fall?”
  - Alternatively: “If two cities are the same in all respects, except that city A has ten more police officers than city B, by how much would the two cities‘ crime rates differ?”
- Experiment:
  - Randomly assign number of police officers to a large number of cities
  - In reality, number of police officers will be determined by crime rate (simultaneous determination of crime and number of police)

Causality and the notion of ceteris paribus

- Effect of the minimum wage on unemployment
  - “By how much (if at all) will unemployment increase if the minimum wage is increased by a certain amount (holding other things fixed)?”
  - What are the potential ‘‘confounding factors‘‘?
- What is the ideal experiment?

Causality and the notion of ceteris paribus

- Effect of the minimum wage on unemployment
  - “By how much (if at all) will unemployment increase if the minimum wage is increased by a certain amount (holding other things fixed)?”
- Experiment:
  - Government randomly chooses minimum wage each year and observes unemployment outcomes
  - Experiment will work because level of minimum wage is unrelated to other factors determining unemployment
  - In reality, the level of the minimum wage will depend on political and economic factors that also influence unemployment

Causality and the notion of ceteris paribus

- Testing predictions of economic theories
  - Economic theories are not always stated in terms of causal effects
  - For example, the expectations hypothesis states that long term interest rates equal compounded expected short term interest rates

$$(1+r_{lt})^n = (1+r^e_{year1})(1+r^e_{year2})\cdot\ldots\cdot(1+r^e_{yearn})$$

  - An implicaton is that the interest rate of a three-months T-bill should be equal to the expected interest rate for the first three months of a six-months T-bill; this can be tested using econometric methods

Reviews

Contents

- What is econometrics?
- Steps in an empirical analysis
- The structure of economic data
- Causality and the notion of Ceteris Paribus

Reading: Chapter 1

Next lecture: Statistics Review (Appendix A, B, C)